

Multilingual Instruction Tuning With Just a Pinch of Multilinguality

Uri Shaham^{τ*} Jonathan Herzig^γ Roei Aharoni^γ
 Idan Szpektor^γ Reut Tsarfaty^γ Matan Eyal^γ

^τ Tel Aviv University
^γ Google Research

Abstract

As instruction-tuned large language models (LLMs) gain global adoption, their ability to follow instructions in multiple languages becomes increasingly crucial. One promising approach is cross-lingual transfer, where a model acquires specific functionality on some language by finetuning on another language. In this work, we investigate how multilinguality during instruction tuning of a multilingual LLM affects instruction-following across languages. We first show that many languages transfer some instruction-following capabilities to other languages from even monolingual tuning. Furthermore, we find that only 40 multilingual examples in an English tuning set substantially improve multilingual instruction-following, both in seen and unseen languages during tuning. In general, we observe that models tuned on multilingual mixtures exhibit comparable or superior performance in several languages compared to monolingually tuned models, despite training on 10x fewer examples in those languages. Finally, we find that increasing the number of languages in the instruction tuning set from 1 to only 2, 3, or 4 increases cross-lingual generalization. Our results suggest that building massively multilingual instruction-tuned models can be done with only a very small set of multilingual instruction-responses.

1 Introduction

Instruction tuning is a fundamental aspect of building modern general-purpose large language models (LLMs), involving fine-tuning a pre-trained model on pairs of instructions and corresponding responses (Mishra et al., 2022; Wei et al., 2022; Sanh et al., 2022; Ouyang et al., 2022). Most instruction tuning datasets are typically limited to English examples, however, for these models to be globally applicable, they must operate on a wide-range of languages.

While creating language-specific datasets from scratch is challenging, cross-lingual transfer has emerged as a promising approach in which a model is finetuned using one language, and acquiring similar abilities in another (Pires et al., 2019; Wu and Dredze, 2019; Artetxe and Schwenk, 2019; K et al., 2020; Conneau et al., 2020a,b). The ability to follow instructions even for languages a model had seen only during pretraining can significantly expand the applicability of LLMs, allowing them to be used by more people that speak different languages. In this work, we show that instruction tuning multilingual LLMs transfers across languages better than previously known, and that even a small amount of language diversity in the tuning set can go a long way for generalizing instruction-following capabilities across languages from the pretraining corpus.

We investigate the effect of instruction tuning cross-lingual transfer using a modern multilingual pretrained LLM and high-quality, open-ended instructions and responses in 12 languages, across different families and writing systems. Notably, tuning on each of the languages individually results in instruction-following capabilities across languages, when English, Italian, and Spanish transfer the most multilingual capabilities on average.

Inspired by this result, we examine the trade-off between a balanced multilingual training set and an English-only one. We find that replacing just 40 English training examples with multilingual examples significantly improves the instruction-following abilities in those languages. Surprisingly, this small amount of language-diverse examples also improves performance for languages that are not represented in the instruction tuning set at all.

We then turn to explore the effect of the number of different languages in the instruction tuning set. We find that having a few languages enable better instruction-following generalization for languages unseen during tuning, comparing to mono-

*Work done during an internship at Google Research.

lingual instruction tuning sets with the same number of examples. Finally, we test two natural potential factors that might influence the degree of cross-lingual transfer, language similarity and the amount of language-specific pretraining data, but find no significant correlations. Overall, our results provide recipes for multilingual instruction tuning that improves cross-lingual generalization, while preserving performance on English, under a fixed budget. In particular, we find that capable multilingual instruction-following models can be tuned even with a minimal amount of multilingual training data.

2 Experimental Setup

We construct our experimental setup with the goal of discovering how multilinguality during instruction tuning affects general-purpose instruction-following models across languages. To that end, we use open-ended instructions and a modern multilingual pretrained LLM. Moreover, we evaluate instruction-following abilities per language in a controlled setting, by using generations of monolingually tuned models per language as a high-bar baseline.

Data We use datasets of high-quality open-ended instructions, rather than classic task-specific ones. Specifically, our training data contains a total of 4,640 English prompts and responses, 1,000 from LIMA (Zhou et al., 2023) and 3,640 from OpenAssistant¹ (Köpf et al., 2023) train sets. Examples from these datasets resemble real world scenarios of users interacting with chatbots with queries like "Can you explain Fermat’s Last Theorem?" and "How to keep a dog hydrated?", and were shown to be effective for instruction tuning LLMs even with a small train set (Zhou et al., 2023). For evaluation, we use 617 prompts from AlpacaFarm (Dubois et al., 2023), originated from Self-Instruct (Wang et al., 2023), Vicuna (Chiang et al., 2023), Koala (Geng et al., 2023), and hh-rlhf (Bai et al., 2022).²

To circumvent the lack of multilingual open-ended instruction tuning datasets, we use machine translation to create multilingual variations of the data. To ensure a high translation quality, we use

Language	Code	Family	Script
Arabic	ar	Afro-Asiatic	Arabic
Chinese	zh	Sino-Tibetan	Chinese
Czech	cs	Indo-European	Latin
English	en	Indo-European	Latin
Estonian	et	Uralic	Latin
Finnish	fi	Uralic	Latin
Hebrew	he	Afro-Asiatic	Hebrew
Hindi	hi	Indo-European	Devanagari
Italian	it	Indo-European	Latin
Russian	ru	Indo-European	Cyrillic
Spanish	es	Indo-European	Latin
Swahili	sw	Niger-Congo	Latin

Table 1: Languages used in our main experiments.

Google Translate API³ to translate the instruction-response pairs of the train set and the instructions of the evaluation set to 11 diverse languages, creating parallel training and evaluation sets in Arabic, Chinese, Czech, English, Estonian, Finnish, Hebrew, Hindi, Italian, Russian, Spanish, and Swahili.⁴ While translated data is different from naturally sourced data for single every language, it allows more control as the data size and content are the same for all languages. Table 1 provides an overview of the languages, their families, scripts, and language codes used throughout this work.

Evaluation We conduct a side-by-side automatic evaluation protocol (Bubeck et al., 2023; Dubois et al., 2023; Dettmers et al., 2023; Gudibande et al., 2023; Zheng et al., 2023), in which an LLM assesses two responses for a single prompt, with the goal of identifying the superior one. We follow common practice (Zheng et al., 2023; Zhang et al., 2023) and present both responses to the model twice, alternating the order of the two responses. Figure 8 in Appendix A shows the exact prompt we use. We define a “win” for a certain response if the judge selects it twice irrespective of the order, and a “tie” if the model selects a different response for each order. We use a discounted-tie (Zhou et al., 2023) scoring method, in which a model receives a score of 1 for a win, 0.5 for a tie, and 0 for a loss. We average the scores of individual instructions to get the score over the evaluation set, and present it as percentages, so a score of 50% means the models performance on the evaluation set are indistinguishable.

¹We focus on single-instruction/single-response interactions so we keep only the first prompt and response from conversations in OpenAssistant (Li et al., 2023).

²We exclude AlpacaFarm’s evaluation prompts from OpenAssistant, as we train using its training set.

³<https://cloud.google.com/translate/docs/reference/api-overview>

⁴Languages were selected from Table 21 in Anil et al. (2023), describing the top-50 languages the model (§2) was pretrained on.

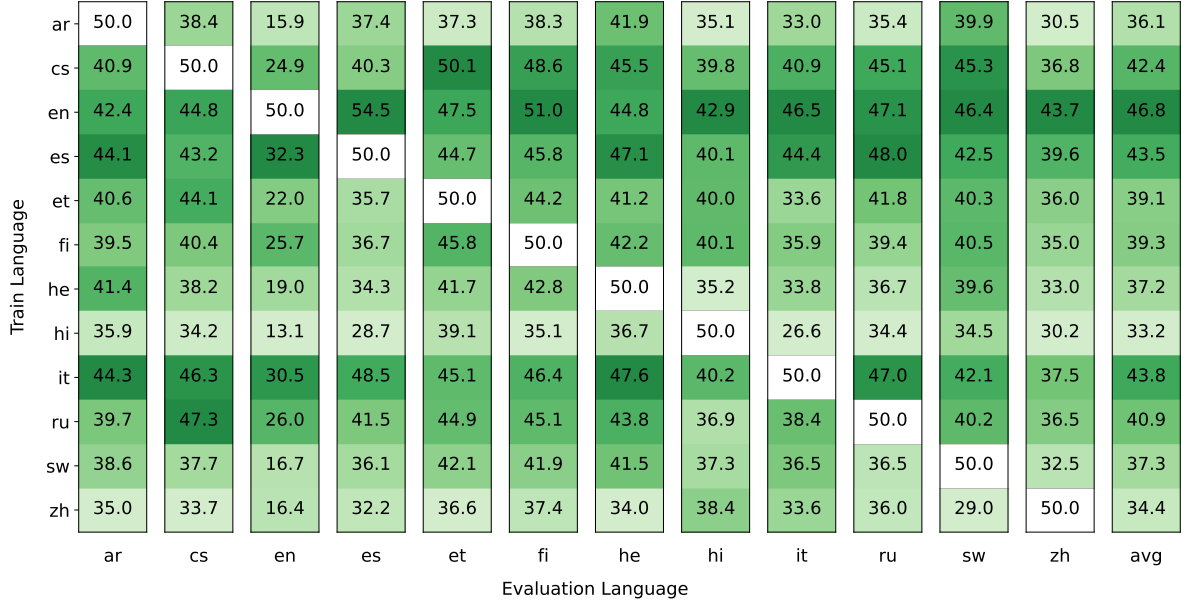


Figure 1: Per language instruction-following scores of models instruction-tuned on monolingual data. Each row represents a model tuned using a different language, and each column is an individual heatmap of the scores of all models on the same evaluation language. Scores are the discounted-ties weighted average of the side-by-side scores against the model tuned on the evaluation language. The scores along the diagonal are 50 as they are the result of comparing generations to themselves, and are excluded from the heatmap coloring.

Instruction-Following Score Per Language

Throughout the paper we measure instruction-following per language by comparing the performance of a model that was finetuned on some training set D , to a model that was monolingually finetuned on the target language $\mathcal{L}$, by using the full training set in this language, $D_{\mathcal{L}}$. We choose this setup as the latter serves as a high bar for the performance on $\mathcal{L}$ using our pretrained LLM, and instruction tuning data. We use the following metric:

$$IT_{\mathcal{L}}(M_D) = S \times S(M_{D_{\mathcal{L}}}, M_D)$$

Where $S \times S(\cdot, \cdot)$ is the side-by-side protocol applied on $M_{D_{\mathcal{L}}}$ and M_D , which are the models instruction-tuned on $D_{\mathcal{L}}$ and D , respectively. A score of 0 means M_D loses on all $\mathcal{L}$ instructions, and 50 means that M_D performance in $\mathcal{L}$ is on the same level as $M_{D_{\mathcal{L}}}$.

Model Throughout our experiments, we use the PaLM-2 models family (Anil et al., 2023) of Transformer (Vaswani et al., 2017) based multilingual pretrained LLMs. We use PaLM 2-S as our pre-trained model for all the instruction tuning experiments, and an instruction-tuned PaLM 2-L as the judge for the side-by-side evaluation. We manually examine the judges decisions and observe it

performs well across languages.

Training and Inference Details We tune every model for 2,000 steps, using a fixed learning rate of $1e-5$, a batch size of 128, and a dropout rate of 0.05. We limit inputs to 1,024 tokens and targets to 512 tokens. We sample a development set of 250 examples from every train set and select the checkpoint based on the development RougeL (Lin, 2004) score. During inference, we generate responses of up to 512 tokens using nucleus sampling (Holtzman et al., 2020) with $p = 0.9$ and temperature of 0.7. For the judge, we use greedy decoding to generate the ID of the better response (1 or 2).

3 How Much Multilinguality Is Needed For Multilingual Instruction Tuning?

We now describe our controlled experiments, designed to quantify the effect of multilingual data during instruction tuning of multilingual LLMs. We start by examining models that are monolingually instruction-tuned, and evaluating their on all languages. We continue with an exploration of the trade-off between different ratios of English and multilingual examples in the tuning set. Lastly, we examine the impact of adding languages to the

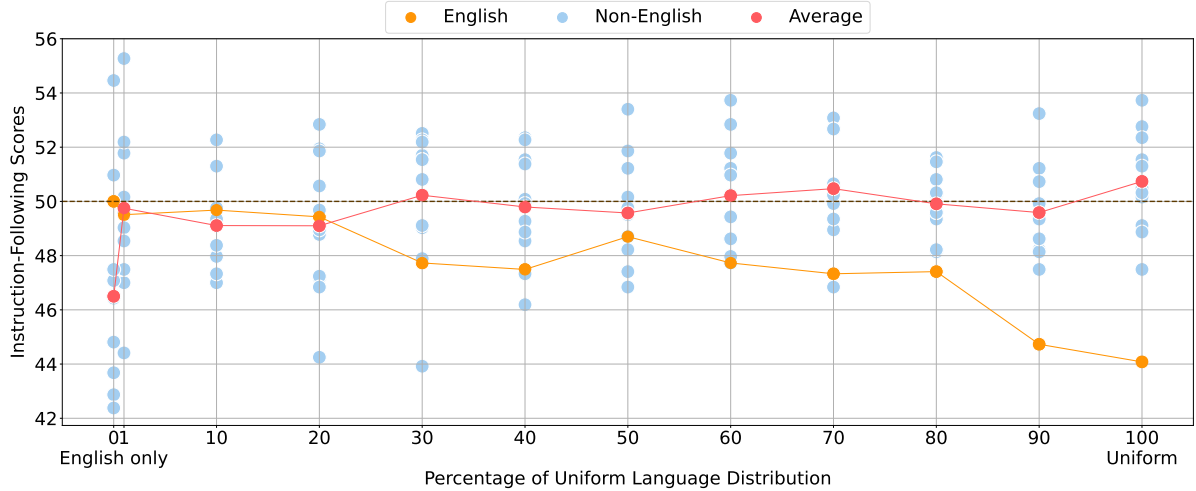


Figure 2: Instruction-following scores of models trained using when $P\%$ of the training set is distributed uniformly across 12 languages and an $(100 - P)\%$ is English only. Each X axis tick represents a tuning mixture, scores over individual non-English languages are in **blue**, and their averages are in **red**. English scores are in **orange**.

tuning set, focusing on how well they generalize to languages unseen during instruction tuning.

3.1 Monolingual Instruction Tuning Yields Multilingual Abilities

To explore zero-shot cross-lingual transfer of instruction tuning in multilingual LLMs, we tune models on a single language and evaluate them on all other languages. We find that all of those models are able to transfer non-negligible instruction-following abilities from their monolingual tuning to other languages.

Setup We instruction-tune 12 models, each one using the full train set translated to a different language. We then prompt each model with the evaluation set in all 12 languages. Finally we calculate their per language scores as described in §2.

Results Figure 1 shows the results, where rows represent training languages and every column is an independent heatmap of the results over a single evaluation language. Most importantly, training using each single language yields a model with multilingual instruction-following capabilities across languages. For context, even the model with the lowest average score, the one tuned on Hindi, achieves a score of over 30 in 9 out of 11 cases. Such a score can be obtained for example by winning 30% of the prompts and losing 70%, or by achieving a tie on 60% of the prompts and losing 40%. The model with the best average score is the one tuned on English, when Italian and Spanish also enable consistently high scores.

Response language When a user prompts a model in a specific language, they usually expect to receive a response in that same language. However, pretrained LLMs are known to often respond in a different language than the language of their prompt (Touvron et al., 2023a; Chen et al., 2023). This also poses a challenge for evaluation of open-ended queries, since those are commonly evaluated using an LLM-as-a-judge (Zheng et al., 2023) protocol, and the judges often ignore whether the response languages match the prompt language, even when instructed not to (Chen et al., 2023). Usually, this is handled by forcing the lowest score to such response (Chen et al., 2023), which does not account for all cases.⁵ To verify our trained models respond in the same language as their prompt, we manually annotate the language of responses to evaluation instructions on all languages. For every language, we randomly sample 20 responses from the pool of models tuned on monolingual data of other languages, to end up with a total of 240 generations from various models. We find that 239 responses are in the same language as the prompt, as desired. This is a major difference in the behavior of our PaLM 2 based instruction-tuned models and the commonly used (Chen et al., 2023) LLaMA-based ones (Touvron et al., 2023a,b). We hypothesize this stems from the multilingual emphasis in the pretraining of PaLM 2, compared to the more English-centric LLaMA.

⁵For example, a response in English to a prompt in French can still be very helpful, or when the prompt is a request for translation or code.

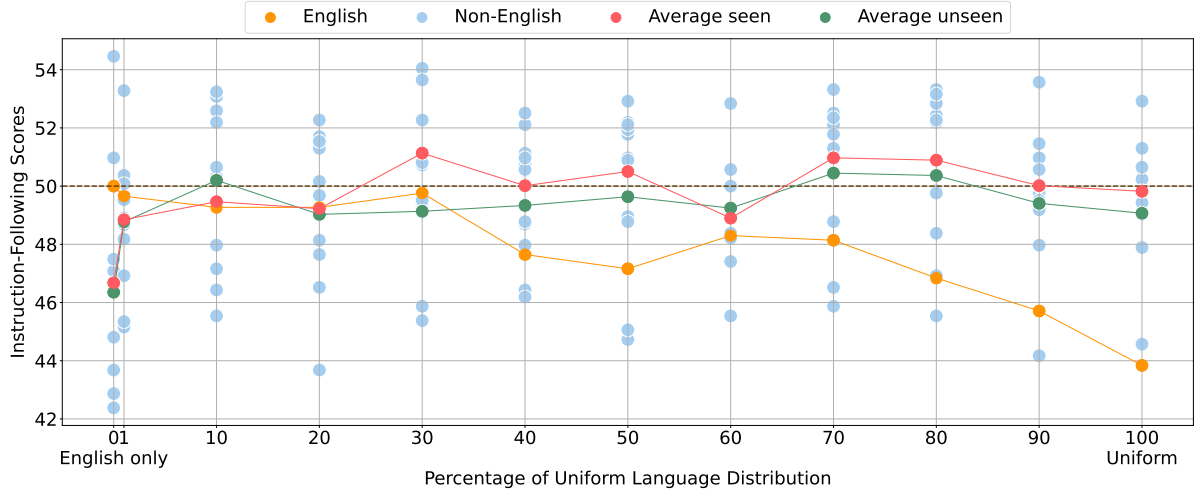


Figure 3: Instruction-following scores of models tuned when $P\%$ of the training set is distributed uniformly across 6 languages and an $(100 - P)\%$ is English only. Each X axis tick represents such a tuning set, scores over individual non-English languages are in blue and English scores are in orange. Average scores of the 5 non-English languages in the tuning set are in red, and the average scores of the 6 languages not seen during tuning are in green.

3.2 A Few Dozen Examples Improve Multilingual Instruction-following

Naturally, multilingual tuning, as opposed to English-exclusive tuning under a fixed budget, should result in better downstream multilingual performance, and might hurt performance on English. Therefore, we ask how many multilingual examples can improve the instruction-following abilities across languages, while preserving English performance. To that end, we train models on data mixtures that include a subset of the English set and subsets of the multilingual set in different ratios. We find a significant boost in multilingual instruction-following abilities even when using only a few dozens of multilingual examples.

Setup We create data mixtures with $P\%$ examples that are evenly split among all 12 languages, and the rest $(100 - P)\%$ English examples.⁶ We create such a train set for every P from 10 to 100, incremented by tens, and also $P = 1$ for which only 40 multilingual examples are included across all 11 non-English languages, and the rest are in English. Finally, we evaluate every tuned model on every one of the 12 languages as defined in §2.

Results Figure 2 visualizes the results. As expected, multilingual examples in the train set improve the score on their languages (Red), and diluting the number of English examples hurts the per-

⁶Every example appears exactly once in every mixture, in a single language.

formance in English (Green). Notably, the significant multilingual improvement comes from replacing only 1% of the English examples by multilingual ones, which translates to 40 examples evenly distributed among the training languages. These results on the effect of such a small amount of language-diversity extend findings regarding task-diversity by Zhou et al. (2023), which demonstrated that a capable monolingual instruction-following model can be tuned using only 1,000 high-quality examples. A second trend is that these models often outperform their monolingually-tuned counterparts on the very language the latter were exclusively tuned on (blue markers above the 50 line). For example, the model tuned using the uniform set ($P = 100$) preforms similarly or better than the individual monolingually-tuned models in 8 of 12 languages, despite being trained on 12 times less instruction-response pairs for each language. This suggests that for some languages, multilingual tuning can enable better instruction-following abilities compared to a traditional monolingual tuning with the same number of examples.

3.3 A Few Dozen Examples Improve Cross-lingual Generalization

Combining the results on zero-shot cross-lingual generalization and the effect of a small amount of multilingual examples from previous sections, we turn to examine how multilingual examples in the tuning set affect language generalization. Specifically, we conduct a similar experiment to the one

in §3.2, this time using only half of the languages for tuning while the rest of languages are unseen. In line with the results from §3.2, we find that a very small amount of multilingual examples also improve performance on languages that were not in the tuning set.

Setup We repeat the setup from §3.2, this time with only English and 5 more languages: Arabic, Finnish, Italian, Russian, and Swahili, and evaluate models again on all 12 languages.

Results Results in Figure 3 show similar trends to the ones in Figure 2. Specifically, the average score over non-English training languages (Red) again improves very quickly, even with $P = 1$. Strikingly, this is also true for languages that the model only seen during pretraining, and are not represented at all in the instruction tuning dataset (Orange). This suggests that very few multilingual examples can not only improve performance for the languages of those examples, but also enable better cross-lingual instruction-following generalization.

3.4 Even a Small Number of Languages Improves Cross-Lingual Generalization

Following the results on the impact of a small number of multilingual examples from various languages, we hypothesize that a small number of languages can also enhance cross-lingual generalization. We experiment with increasing the number of languages in the tuning set and observe that cross-lingual transfer to languages only seen during pretraining improves from the very first additional languages.

Setup We instruction-tune models on a single language and up to 6 languages, with each incremental step adding a language to the tuning set. We keep the total number of examples fixed and split them uniformly among the languages. We use the 6 training languages from Section 3.3, and follow three different permutations that determine the order in which we add languages to the mix. These permutations are shown in Table 3 in Appendix B. We evaluate every model on each of the remaining 6 languages with the same procedure as the previous experiments, and average scores per such evaluation language across models that are tuned using the same number of languages.

Results Results on Figure 4 show that adding languages to the tuning set improves cross-lingual

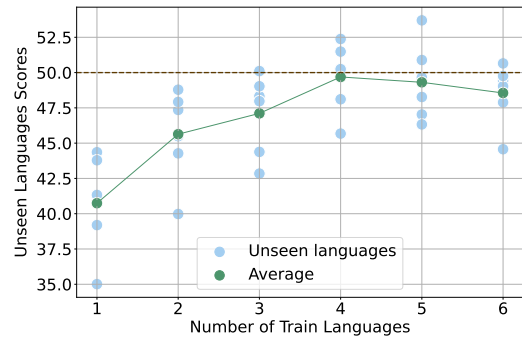


Figure 4: Instruction-following scores on Czech, Estonian, Hebrew, Hindi, Spanish, and Chinese of models instruction-tuned using various subsets of Arabic, English, Finnish, Italian, Russian, and Swahili. Blue markers are the average scores per evaluation languages across models tuned with the same number of languages. The averages of those individual languages scores are in green.

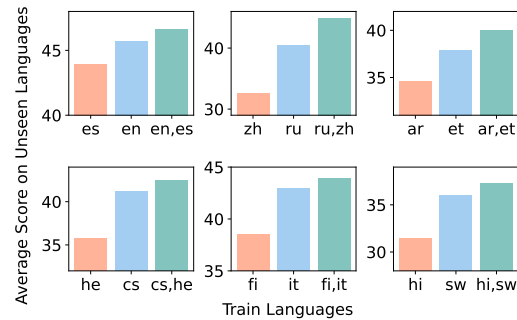


Figure 5: Average instruction-following scores of languages not seen during instruction tuning. For example, the top-left corner describes the scores of 3 models instruction-tuned on 100% Spanish, 100% English, and 50% Spanish and 50% English. The Y axis of this top-left sub-figure is the average score across all language excluding Spanish and English.

generalization. The average score (Red) increases from tuning on monolingual data to tuning on bilingual data, and even more when using 3 and 4 languages, where the average score almost gets to 50. At that point, there is an indication for saturation, as adding more languages does not seem to improve transfer further. These findings demonstrate that diversifying the instruction tuning data with only a few different languages can improve cross-lingual transfer to new languages, only seen during pretraining.

Bilingual Tuning Sets To show this phenomenon holds for even more combinations of different lan-

guages we randomly split all 12 languages to pairs, and train models using 50% of the examples in the one language and 50% in the other one. We evaluate each of these models on the remaining 10 languages, and compare their average score to the average scores of the two models tuned using the full monolingual sets. Results on Figure 5 reveal that instruction tuning using two languages helps generalize better to languages not seen during instruction tuning, as these models achieve better average scores across the board.

4 Potential Factors of Cross-Lingual Transfer

Following the results above, a natural question arises: what factors predict the degree of cross-lingual transfer? We explore two immediate candidates. Initially, we investigate the impact of various aspects of language similarity between a pair of languages on cross-lingual transfer between them. Subsequently, we examine whether the proportion of language-specific data utilized in pretraining correlates with the average cross-lingual transfer efficacy during instruction tuning using the given language.

4.1 Language Similarity

A intuitive hypothesis is that aspects of language similarity like the script or mutual intelligibility might affect the levels of instruction tuning cross-lingual transfer between languages. We test this using a case study of 4 Slavic languages, looking into possible effects of such aspects. However, we do not find a signal indicating these factors strongly correlate with cross-lingual transfer.

Setup We train models on a monolingual version of the data in Russian, Serbian, Croatian, and Slovenian, and evaluate their transfer to each other. We select those 4 languages as they can be divided along several interesting linguistic lines. First, Russian is East Slavic, and the rest are South Slavic. Second, Russian and Serbian use the Cyrillic script, while Croatian and Slovenian use Latin. Moreover, Serbian and Croatian share a significant degree of mutual intelligibility. These difference are summarized in Table 2.

Results Results are displayed on Figure 6. As shown, there is not a strong signal indicating any of the aspects above is related with better mutual cross-lingual transfer. Specifically, Russian tend to

Language	Code	Slavic Family	Script	Mutually Intelligible
Russian	ru	East	Cyrillic	-
Serbian	sr	South	Cyrillic	Croatian
Croatian	hr	South	Latin	Serbian
Slovenian	sl	South	Latin	-

Table 2: Languages used for language similarity experiment, along with their language code, subfamily, script, and the language they are mutually intelligible with.

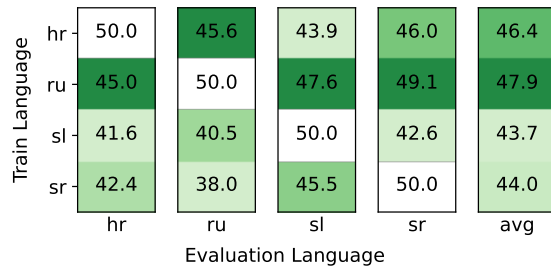


Figure 6: Instruction-following scores per language of models tuned on monolingual instruction tuning set. Each row represents a model trained using a different language, and each column is an individual heatmap of the scores of all models on the same evaluation language. The scores along the diagonal are excluded from the heatmap coloring

best transfer instruction-following abilities to rest, and while Russian and Serbian both use Cyrillic, it is Croatian that transfers capabilities to Russian significantly better in our study. Moreover, Despite being largely mutual intelligible, Croatian and Serbian do not seem to share cross-lingual abilities more than the others. Our results align with recent findings that language similarity does not impact transferability or interference in machine translation given sufficient data and model capacity (Fernandes et al., 2023; Shaham et al., 2023).

4.2 Fraction of Data in Pretraining

A second possible factor influencing the degree of cross-lingual transfer from a particular language is the extent to which the model was exposed to it during pretraining. Generally, a model’s downstream performance on a specific language correlates with the fraction of data in that language in the pretraining corpus (Muennighoff et al., 2023). In contrast, Figure 7 suggests this is not necessarily the case for the cross-lingual transfer from a specific language. We find a weak correlation between the average cross-lingual score of each language and the number of documents in that language in the models pretraining corpus (Table 21 in Anil et al. (2023)).

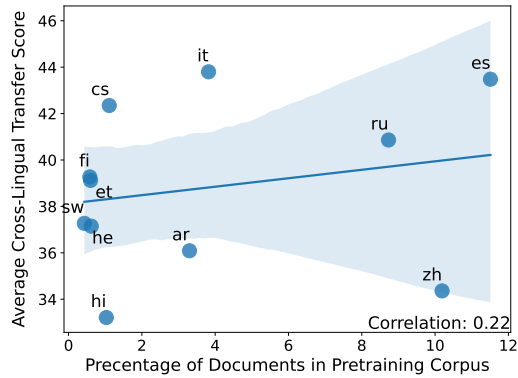


Figure 7: Weak Pearson correlation between the percentage of documents in the pretraining corpus (excluding English), and the average instruction-following score across languages for every single training language. Blue area surrounding the line is the confidence interval.

5 Related work

Cross-lingual Transfer The success of the pretrain-in-finetuning paradigm (Devlin et al., 2019) ignited a new line of work on cross-lingual transfer. Pires et al. (2019) and Wu and Dredze (2019) showed that the multilingual variant of BERT (Devlin et al., 2019), Multilingual-BERT (mBERT) can be finetuned on a specific task in one language and preform this task on another language. Artetxe and Schwenk (2019) reported similar findings with a Recurrent Neural Network on an English natural language inference dataset. K et al. (2020) explored how different factors like linguistic similarity and model architecture impact cross-lingual transfer. Conneau et al. (2020a) introduced XLM-R, a multilingual pretrained encoder with superior cross-lingual performance to mBERT, and Phang et al. (2020) showed that intermediate training on an English task after pretraining improves XLM-R’s cross-lingual transfer further. Conneau et al. (2020b) demonstrated that parameter sharing is more important for cross-lingual transfer than shared vocabulary. Malkin et al. (2022) investigated how pretraing BERT-based models on different language pairs affect cross-lingual downstream performance. Going beyond encoder-only models, Xue et al. (2021) proposed mT5, a multilingual variant of T5 (Raffel et al., 2020), and showed the significance of model scaling for cross-lingual transfer abilities by finetuning on generative tasks in English, and evaluating on other languages. While a vast amount of cross-lingual transfer lit-

erature is focused on task-specific finetuning, we explore trends of transfer for a general-purpose instruction-following LLMs.

Multilingual Instruction Tuning Naturally, initial work on instruction tuning (Mishra et al., 2022; Wei et al., 2022; Sanh et al., 2022) focused on cross-tasks generalization in English. Following these, a large body of work was dedicated to multilingual instruction tuning. Muennighoff et al. (2023) found that tuning models with English datasets enables zero-shot cross-lingual abilities to new languages. The authors also found this holds for languages that the model never intentionally seen during pretraining, and that multilingual training improves generalization across tasks. Ye et al. (2023) explored cross-lingual transfer with an emphasis on English-centric models with LLaMA (Touvron et al., 2023a). Chen et al. (2023) investigated the effects of full parameter training vs low-rank adaptation (Hu et al., 2022) and monolingual vs multilingual instruction tuning using the Stanford Alpaca project (Taori et al., 2023) data, machine translated into five languages. Lai et al. (2023) trained multilingual instruction-following models for 26 languages with reinforcement learning from human feedback (Ouyang et al., 2022), and Zhang et al. (2023) suggested instruction tuning LLMs by prepending the instruction and response translated into a pivot language (e.g English) to the response in the target language. In this work, we consider cross-lingual transfer from twelve languages, rather than focusing on the transferability achieved by English tuning, and demonstrate instruction tuning trends for an LLM pretrained with a significant emphasis on multilinguality, which unlocks consistent cross-lingual performance. Importantly, we unveil the potential of just a small amount of language diversity in the instruction tuning set for cross-lingual generalization.

6 Conclusion

This work demonstrates that cross-lingual transfer offers a promising avenue for building massively multilingual instruction-following LLMs. Our findings suggest that monolingual instruction tuning using many different training languages can result in instruction-following capabilities in many other languages. Moreover, incorporating even a small set of multilingual examples (as little as 40) can significantly enhance instruction-following performance in both the languages the model is tuned

on, and ones that were only seen during pretraining. Additionally, for some languages, training on such multilingual datasets achieves comparable or even superior performance compared to monolingual instruction tuning. We observe a similar trend when exploring the effect of total number of languages in the tuning set, as even splitting the train set to only two languages improves generalization to new languages, comparing to monolingual tuning. These findings pave the way for efficient and scalable development of multilingual LLMs capable of understanding and following instructions across diverse languages with minimal multilingual data requirements.

7 Limitations

Limitations of our work include the use of translation for expanding datasets to multilingual settings, the number of languages we evaluated on, and number of models we experimented with. We now discuss each of them.

Translated data One limitation of our work is that our data is translated using Google Translate API, and not originally sourced by native speakers. Automatic translation is inherently imperfect and may introduce noise to the tuning sets. However, translation also allows to for a controlled setup, in which the content of all training and evaluation examples is the same for all languages.

Number of languages A second limitation is that we use 12 languages in our main experiments (§3), with 3 additional languages in the language similarity experiment (§4.1). Clearly, multilingual instruction-following models need to be able to operate in more languages, and we leave work on scaling this number to future work.

Number of models Lastly, we experiment with PaLM 2, and results may vary with different LLMs. Nevertheless, our focus on PaLM 2 highlights the potential of multilingual pretraining for future advancements in LLMs.

References

Rohan Anil, Andrew M. Dai, Orhan Firat, Melvin Johnson, Dmitry Lepikhin, Alexandre Passos, Siamak Shakeri, Emanuel Taropa, Paige Bailey, Zhifeng Chen, Eric Chu, Jonathan H. Clark, Laurent El Shafey, Yanping Huang, Kathy Meier-Hellstern, Gaurav Mishra, Erica Moreira, Mark Omernick, Kevin Robinson, Sebastian Ruder, Yi Tay, Kefan Xiao,

Yuanzhong Xu, Yujing Zhang, Gustavo Hernandez Abrego, Junwhan Ahn, Jacob Austin, Paul Barham, Jan Botha, James Bradbury, Siddhartha Brahma, Kevin Brooks, Michele Catasta, Yong Cheng, Colin Cherry, Christopher A. Choquette-Choo, Aakanksha Chowdhery, Clément Crepy, Shachi Dave, Mostafa Dehghani, Sunipa Dev, Jacob Devlin, Mark Díaz, Nan Du, Ethan Dyer, Vlad Feinberg, Fangxiaoyu Feng, Vlad Fienber, Markus Freitag, Xavier Garcia, Sebastian Gehrmann, Lucas Gonzalez, Guy Gur-Ari, Steven Hand, Hadi Hashemi, Le Hou, Joshua Howland, Andrea Hu, Jeffrey Hui, Jeremy Hurwitz, Michael Isard, Abe Ittycheriah, Matthew Jagielski, Wenhao Jia, Kathleen Kenealy, Maxim Krikun, Sneha Kudugunta, Chang Lan, Katherine Lee, Benjamin Lee, Eric Li, Music Li, Wei Li, YaGuang Li, Jian Li, Hyeontaek Lim, Hanzhao Lin, Zhongtao Liu, Frederick Liu, Marcello Maggioni, Aroma Mahendru, Joshua Maynez, Vedant Misra, Maysam Moussalem, Zachary Nado, John Nham, Eric Ni, Andrew Nystrom, Alicia Parrish, Marie Pellat, Martin Polacek, Alex Polozov, Reiner Pope, Siyuan Qiao, Emily Reif, Bryan Richter, Parker Riley, Alex Castro Ros, Aurko Roy, Brennan Saeta, Rajkumar Samuel, Renee Shelby, Ambrose Slone, Daniel Smilkov, David R. So, Daniel Sohn, Simon Tokumine, Dasha Valter, Vijay Vasudevan, Kiran Vodrahalli, Xuezhi Wang, Pidong Wang, Zirui Wang, Tao Wang, John Wieting, Yuhuai Wu, Kelvin Xu, Yunhan Xu, Linting Xue, Pengcheng Yin, Jiahui Yu, Qiao Zhang, Steven Zheng, Ce Zheng, Weikang Zhou, Denny Zhou, Slav Petrov, and Yonghui Wu. 2023. [PaLM 2 technical report](#).

Mikel Artetxe and Holger Schwenk. 2019. [Massively multilingual sentence embeddings for zero-shot cross-lingual transfer and beyond](#). *Transactions of the Association for Computational Linguistics*, 7:597–610.

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt, Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, Ben Mann, and Jared Kaplan. 2022. [Training a helpful and harmless assistant with reinforcement learning from human feedback](#).

Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrmann, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, Harsha Nori, Hamid Palangi, Marco Tulio Ribeiro, and Yi Zhang. 2023. [Sparks of artificial general intelligence: Early experiments with gpt-4](#).

Pinzhen Chen, Shaoxiong Ji, Nikolay Bogoychev, Barry Haddow, and Kenneth Heafield. 2023. [Monolingual or multilingual instruction tuning: Which makes a better alpaca](#).

- Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E. Gonzalez, Ion Stoica, and Eric P. Xing. 2023. [Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality](#).
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020a. [Unsupervised cross-lingual representation learning at scale](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.
- Alexis Conneau, Shijie Wu, Haoran Li, Luke Zettlemoyer, and Veselin Stoyanov. 2020b. [Emerging cross-lingual structure in pretrained language models](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6022–6034, Online. Association for Computational Linguistics.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. [QLoRA: Efficient finetuning of quantized LLMs](#). In *Thirty-seventh Conference on Neural Information Processing Systems*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Yann Dubois, Xuechen Li, Rohan Taori, Tianyi Zhang, Ishaan Gulrajani, Jimmy Ba, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. [Alpaca-farm: A simulation framework for methods that learn from human feedback](#).
- Patrick Fernandes, Behrooz Ghorbani, Xavier Garcia, Markus Freitag, and Orhan Firat. 2023. [Scaling laws for multilingual neural machine translation](#).
- Xinyang Geng, Arnav Gudibande, Hao Liu, Eric Wallace, Pieter Abbeel, Sergey Levine, and Dawn Song. 2023. [Koala: A dialogue model for academic research](#). Blog post.
- Arnav Gudibande, Eric Wallace, Charlie Snell, Xinyang Geng, Hao Liu, Pieter Abbeel, Sergey Levine, and Dawn Song. 2023. [The false promise of imitating proprietary llms](#).
- Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. [The curious case of neural text de-generation](#). In *International Conference on Learning Representations*.
- Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [LoRA: Low-rank adaptation of large language models](#). In *International Conference on Learning Representations*.
- Karthikeyan K, Zihan Wang, Stephen Mayhew, and Dan Roth. 2020. [Cross-lingual ability of multilingual bert: An empirical study](#). In *International Conference on Learning Representations*.
- Andreas Köpf, Yannic Kilcher, Dimitri von Rütte, Sotiris Anagnostidis, Zhi-Rui Tam, Keith Stevens, Abdullah Barhoum, Nguyen Minh Duc, Oliver Stanley, Richárd Nagyfi, Shahul ES, Sameer Suri, David Glushkov, Arnav Dantuluri, Andrew Maguire, Christoph Schuhmann, Huu Nguyen, and Alexander Mattick. 2023. [Openassistant conversations – democratizing large language model alignment](#).
- Viet Lai, Chien Nguyen, Nghia Ngo, Thuat Nguyen, Franck Dernoncourt, Ryan Rossi, and Thien Nguyen. 2023. [Okapi: Instruction-tuned large language models in multiple languages with reinforcement learning from human feedback](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 318–327, Singapore. Association for Computational Linguistics.
- Xian Li, Ping Yu, Chunting Zhou, Timo Schick, Luke Zettlemoyer, Omer Levy, Jason Weston, and Mike Lewis. 2023. [Self-alignment with instruction back-translation](#).
- Chin-Yew Lin. 2004. [ROUGE: A package for automatic evaluation of summaries](#). In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain. Association for Computational Linguistics.
- Dan Malkin, Tomasz Limisiewicz, and Gabriel Stanovsky. 2022. [A balanced data approach for evaluating cross-lingual transfer: Mapping the linguistic blood bank](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4903–4915, Seattle, United States. Association for Computational Linguistics.
- Swaroop Mishra, Daniel Khashabi, Chitta Baral, and Hannaneh Hajishirzi. 2022. [Cross-task generalization via natural language crowdsourcing instructions](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3470–3487, Dublin, Ireland. Association for Computational Linguistics.
- Niklas Muennighoff, Thomas Wang, Lintang Sutawika, Adam Roberts, Stella Biderman, Teven Le Scao, M Saiful Bari, Sheng Shen, Zheng Xin Yong, Hailey Schoelkopf, Xiangru Tang, Dragomir Radev, Alham Fikri Aji, Khalid Almubarak, Samuel Albanie, Zaid Alyafeai, Albert Webson, Edward Raff, and Colin Raffel. 2023. [Crosslingual generalization through multitask finetuning](#). In *Proceedings*

- of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 15991–16111, Toronto, Canada. Association for Computational Linguistics.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F Christiano, Jan Leike, and Ryan Lowe. 2022. [Training language models to follow instructions with human feedback](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744. Curran Associates, Inc.
- Jason Phang, Iacer Calixto, Phu Mon Htut, Yada Puk-sachatkun, Haokun Liu, Clara Vania, Katharina Kann, and Samuel R. Bowman. 2020. [English intermediate-task training improves zero-shot cross-lingual transfer too](#). In *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing*, pages 557–575, Suzhou, China. Association for Computational Linguistics.
- Telmo Pires, Eva Schlinger, and Dan Garrette. 2019. [How multilingual is multilingual BERT?](#) In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4996–5001, Florence, Italy. Association for Computational Linguistics.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. [Exploring the limits of transfer learning with a unified text-to-text transformer](#). *Journal of Machine Learning Research*, 21(140):1–67.
- Victor Sanh, Albert Webson, Colin Raffel, Stephen Bach, Lintang Sutawika, Zaid Alyafeai, Antoine Chaffin, Arnaud Stiegler, Arun Raja, Manan Dey, M Saiful Bari, Canwen Xu, Urmish Thakker, Shanya Sharma Sharma, Eliza Szczechla, Taewoon Kim, Gunjan Chhablani, Nihal Nayak, Debajyoti Datta, Jonathan Chang, Mike Tian-Jian Jiang, Han Wang, Matteo Manica, Sheng Shen, Zheng Xin Yong, Harshit Pandey, Rachel Bawden, Thomas Wang, Trishala Neeraj, Jos Rozen, Abheesht Sharma, Andrea Santilli, Thibault Fevry, Jason Alan Fries, Ryan Teehan, Teven Le Scao, Stella Biderman, Leo Gao, Thomas Wolf, and Alexander M Rush. 2022. [Multi-task prompted training enables zero-shot task generalization](#). In *International Conference on Learning Representations*.
- Uri Shaham, Maha Elbayad, Vedanuj Goswami, Omer Levy, and Shruti Bhosale. 2023. [Causes and cures for interference in multilingual translation](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15849–15863, Toronto, Canada. Association for Computational Linguistics.
- Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. [Stanford alpaca: An instruction-following llama model](#). https://github.com/tatsu-lab/stanford_alpaca.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023a. [Llama: Open and efficient foundation language models](#).
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurelien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. 2023b. [Llama 2: Open foundation and fine-tuned chat models](#).
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. [Attention is all you need](#). In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.
- Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. [Self-instruct: Aligning language models with self-generated instructions](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13484–13508, Toronto, Canada. Association for Computational Linguistics.
- Jason Wei, Maarten Bosma, Vincent Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V Le. 2022. [Finetuned language models are zero-shot learners](#). In *International Conference on Learning Representations*.
- Shijie Wu and Mark Dredze. 2019. [Beto, bentz, becas: The surprising cross-lingual effectiveness of BERT](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 833–844, Hong

Kong, China. Association for Computational Linguistics.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. [mT5: A massively multilingual pre-trained text-to-text transformer](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 483–498, Online. Association for Computational Linguistics.

Jiacheng Ye, Xijia Tao, and Lingpeng Kong. 2023. [Language versatilitists vs. specialists: An empirical revisiting on multilingual transfer ability](#).

Zhihan Zhang, Dong-Ho Lee, Yuwei Fang, Wenhao Yu, Mengzhao Jia, Meng Jiang, and Francesco Barbieri. 2023. [Plug: Leveraging pivot language in cross-lingual instruction tuning](#).

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric. P Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. [Judging llm-as-a-judge with mt-bench and chatbot arena](#).

Chunting Zhou, Pengfei Liu, Puxin Xu, Srini Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. 2023. [Lima: Less is more for alignment](#).

A Side-by-side evaluation

Figure 8 shows the prompt given the the LLM judge for the side-by-side evaluation.

Below is an instruction and two answers. Choose your preferred answer, which can be subjective.

The instruction:

{instruction}

Answer1:

{response 1}

Answer2:

{response 2}

Which one is better, Answer1 or Answer2?

Only write a single digit as your answer, '1' for Answer1 or '2' for Answer2. Do not add any explanation.

Figure 8: Side-by-side evaluation prompt.

B Languages Permutations

Table 3 displays the 3 permutations of 6 languages. We use these permutations to determine the order in which we add languages to the tuning set in the experiment described Section 3.4.

1	2	3	4	5	6
fi	fi,en	fi,en,ru	fi,en,ru,it	fi,en,ru,it,sw	all six
sw	sw,it	sw,it,ar	sw,it,ar,en	sw,it,ar,en,fi	all six
it	it,fi	it,fi,en	it,fi,en,ar	it,fi,en,ar,ru	all six

Table 3: Subsets of languages used to tune models for the experiment described in Section 3.4. Each cell represents a version of the training set, for which all examples are uniformly split between the languages in that cell. Evaluation score for language $\mathcal{L}$ is calculated as the average score on $\mathcal{L}$ for all models within a specific column.